Un iversity: **University of Eloued**
Faculty: **Faculty of Economic, Commercial and Management Sciences**
Instructor: **Dr. Atmane Medini**
Academic Year: **2025 / 2026**

Time Series Exercises: Stationarity Tests using R

Below is a **structured series of exercises for Time Series Stationarity Analysis using R**, inspired by the workflow used in your repository. The exercises cover **visual inspection, log transformation, differencing, and stationarity tests (ADF, PP, KPSS)**.

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Dataset Variables:

Date
GDP_Index
Inflation_Rate
Interest_Rate
Exchange_Rate
Stock_Market_Index

Exercise 1: Data Preparation and Time Series Construction

Using the dataset provided, perform the following tasks.

1. Import the dataset into R from a CSV file.
2. Convert the variable **Date** into a date format.
3. Transform the dataset into a **time series object**.
4. Display the first observations of the dataset.
5. Plot each variable over time.

Questions

1. What general patterns do you observe in the plots?
2. Do the series appear to have trends or structural movements?
3. Which variables appear more volatile?

Exercise 2: Visual Inspection of Stationarity

For each variable:

GDP_Index
Inflation_Rate
Interest_Rate

Exchange_Rate
Stock_Market_Index

perform the following tasks.

1. Plot the time series.
2. Compute summary statistics.
3. Plot rolling mean and rolling variance (if possible).

Questions

1. Does the mean appear constant over time?
2. Does the variance appear stable?
3. Which variables appear non-stationary?

Exercise 3: Logarithmic Transformation

Many economic time series exhibit **exponential growth**, which can lead to increasing variance.

Tasks

1. Apply the **log transformation** to the following variables:

GDP_Index
Exchange_Rate
Stock_Market_Index

2. Plot the transformed series.
3. Compare the original series with the log-transformed series.

Questions

1. What is the purpose of the logarithmic transformation?
2. Does the variance appear more stable after the transformation?
3. Which series benefits the most from the log transformation?

Exercise 4: First Difference Transformation

To remove trends and achieve stationarity, the first difference is often applied.

Tasks

1. Compute the first difference of the following variables:

GDP_Index
Inflation_Rate
Interest_Rate

Exchange_Rate
Stock_Market_Index

2. Plot the differenced series.
3. Compare the level series and the differenced series.

Questions

1. What effect does differencing have on the trend?
2. Does the mean appear more stable after differencing?
3. Which variables appear stationary after the first difference?

Exercise 5: Augmented Dickey–Fuller (ADF) Test

The **ADF test** checks whether a time series contains a **unit root**, indicating non-stationarity.

Tasks

1. Apply the ADF test to the following series:

GDP_Index
Inflation_Rate
Interest_Rate
Exchange_Rate
Stock_Market_Index

2. Record the **test statistic and p-value**.

Questions

1. What is the null hypothesis of the ADF test?
2. At the 5% significance level, which series are stationary?
3. Which variables require transformation?

Exercise 6: Phillips–Perron (PP) Test

The **Phillips–Perron test** is another unit root test that corrects for serial correlation and heteroskedasticity.

Tasks

1. Apply the PP test to all five variables.
2. Compare the results with the ADF test.

Questions

1. Do the results differ from the ADF test?
2. Which variables appear non-stationary according to both tests?

Exercise 7: KPSS Stationarity Test

Unlike the ADF and PP tests, the **KPSS test assumes that the series is stationary under the null hypothesis**.

Tasks

1. Apply the KPSS test to all variables.
2. Compare the results with the ADF and PP tests.

Questions

1. What is the null hypothesis of the KPSS test?
2. Do the KPSS results confirm the ADF and PP results?
3. What conclusions can be drawn about the stationarity of each variable?

Exercise 8: Stationarity after Transformation

Using the transformed series:

Log(GDP_Index)
Log(Exchange_Rate)
Log(Stock_Market_Index)
First differences of all variables

Tasks

1. Apply the **ADF test** again.
2. Apply the **PP test** again.
3. Apply the **KPSS test** again.

Questions

1. Are the transformed series stationary?
2. Which transformation was most effective?
3. What transformation is required before estimating econometric models such as ARIMA or VAR?

Exercise 9: Interpretation and Economic Discussion

Using the results obtained in the previous exercises:

1. Summarize the stationarity properties of each variable.
2. Identify which variables are **I(0)** and which are **I(1)**.
3. Explain why stationarity is important in time series econometrics.
4. Discuss how non-stationary data may affect regression results.